



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Mid Term Examinations (Spring 2023)

Course Code with Title	CS-229T : Database Systems		Program	Bachelor of Science BS(CS)
Instructor	Mr. Shardha Nand/Dr. Anam Akber		Semester	4 th
Start date & Time	May 20, 2023 at 9:30 AM	End Time	11:00 AM	
Maximum Marks	30			

IMPORTANT INSTRUCTIONS:

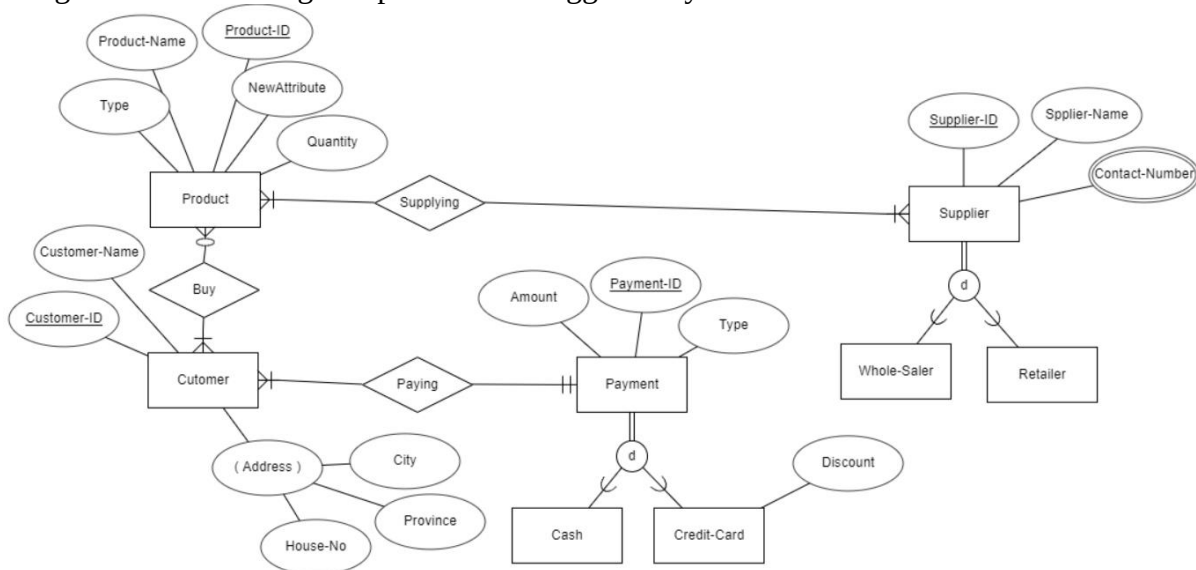
- Attempt All Questions.
- Mobile Phones are strictly prohibited in examination premises.

Q.1.

(05)

A manufacturing company produces several products. The following product information is stored: product name, product ID, type, and quantity on hand. Each product may be supplied by one or more suppliers. There are two types of supplier's i.e. (i) wholesaler (ii) retailer. Supplier has SuppID, SuppName, ProductType and keeping more than two contactNumbers. Customers are buying products. Customer is keeping attributes like CustomerID, Name, and Address (City, Province, HouseNo). Payment may be made through debit card or cash. Payment through debit card is availing discounts. PaymentID, amount, type)

Design an ERD showing all operations of suggested system.





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Q.2.

(03+03)

(a) What are the main issues in below mentioned scenario considering 2nd normal form? Come up with proper design of 2nd NF.

Customer

CustomerID	StoreID	Location
1	1	Gulshan-e-Iqbal
1	3	Clifton
2	1	Gulshan-e-Iqbal
3	2	Saddar
4	3	Clifton
5	2	Saddar
6	3	Clifton

Solution:

It is pointed out that Location is partially dependent on StoreID which violates 2nd NF so schema will be as under.

Customer-Detail	
<u>CustomerID</u>	<u>StoreID</u>
1	1
1	3
2	1
3	2
4	3
5	2
6	3

Store-Information	
<u>StoreID</u>	<u>Location</u>
1	Gulshan-e-Iqbal
2	Saddar
3	Clifton

(b) Why below mentioned solution is not best suited. How would you convert following into 3rd normal form?

<u>Roll No</u>	<u>Province</u>	<u>City</u>
01	Sindh	Karachi
02	Punjab	Lahore
03	Baluchistan	Quetta
04	Khyber Pakhtunikhwa	Peshawar

Solution

This example shows transitive dependency.

Roll_No -> Province

Province -> City

Roll_No -> city So Roll_No shows transitive dependency on city.



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IT shows that non-prime attribute City is identified by another non-prime attribute i.e. Province

Student-Info	
Roll No	City
01	Karachi
02	Lahore
03	Quetta
04	Peshawar

Address	
Province	City
Sindh	Karachi
Punjab	Lahore
Baluchistan	Quetta
Khyber Pakhtunikhwa	Peshawar

Q.3.

(05)

Users	
UserID	PK auto generated
Username	Varchar(12)
Password	Varchar(8)
FirstName	Varchar(25)
LastName	Varchar(25)
Address	Varchar(25)
PhoneNo	Int
City	Varchar(25)
Email	Varchar(30)

Order	
OrderID	PK autogenerated
UserID	FK
ProductID	Varchar(12)
ProductQuantity	Int
ProductCategory	Varchar(12)
OrderEmail	Varchar(12)
OrderAddress	Varchar(50)
OrderPhoneNo	Int
OrderStatus	Varchar(12)

Write queries to

1. Select First name & last name as a new column from User table.
2. Group all Products according to their Category
3. Delete all the Orders from table whose status is "Not Available"
4. Alter User table by adding Order History Column
5. Select category for all orders where product quantity is above 100.

Solution

1. **Select First name & last name as a new column from User table.**
 Select concat(FirstName,LastName) AS Name from Users
2. **Group all Products according to their Category**
 Select ProductID from Order Group By ProductCategory'
3. **Delete all the Orders from table whose status is "Not Available"**
 Delete * from Order where OrderStatus = "Not Available"
4. **Alter User table by adding Order History Column**
 Alter Order add OrderHistory
5. **Select category for all orders where product quantity is above 100.**
 Select ProductCategory from order where Count(ProductQuantity) > 100



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Q.4.**(05)**

Functional dependencies are a fundamental concept in database design, and are used to establish relationships between attributes in a database. Considering this important aspect, find out (i) functional dependencies (ii) Super keys, Candidate keys, Primary Key and alternate keys considering below mentioned table.

StudentID	Name	Department	Email
1	Saleemullah	CS	Saleemullah@gmail.com
2	Asad Ali	IT	Asad.Ali@yahoo.com
3	Karamat Khan	CS	Karamat-Khan@hotmail.com
4	Sultan Ahmed	IT	Sultan.Ahmed@gmail.com
5	Muhammad Javed	CS	Muhammad_Javed@yahoo.com

Solution**Functional dependency**

Student-ID	→	Email
Student-ID	→	Name
Student-ID	→	Department
Student-ID, Dept	→	Name
Student-ID, Name, Dept	→	Email
Student-ID, Email	→	Department
Email	→	Student-ID
Email, Name	→	Student-ID

Super Keys

Student-ID	Email	Student-ID, Dept	Student-ID, Email
Student-ID, Name, Dept	Email, Name	Email, Dept	Student-ID, Name, Dept, Email

Candidate Key

Student-ID	Email
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Primary Key	Student-ID
Alternate Key	Email-ID



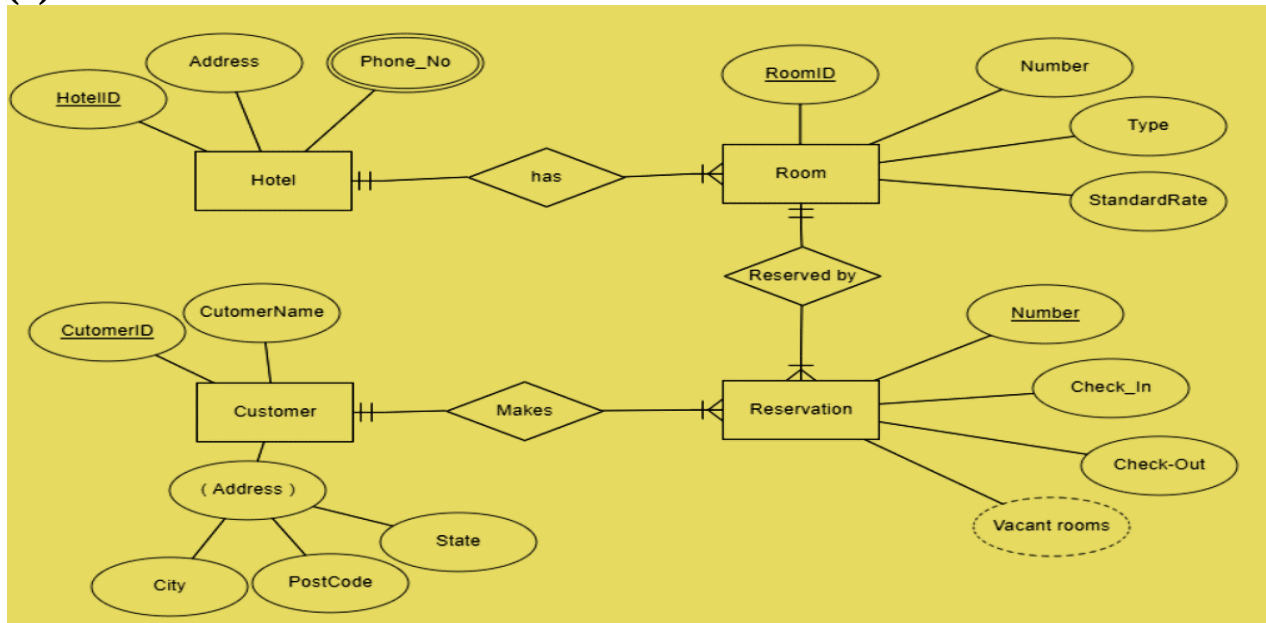
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Q.5.

(05+04)

Relational Schema shows how the companies store and organize information. Being a database designer, you are required to efficiently map following ERD/Enhanced ERD into relational model.

(a)



Solution

Hotel (Hotel-ID, Address)

Phone(Phone-No, Hotel-ID)

Room(Room-ID, Number, type, Standard-Rate, Hotel-ID)

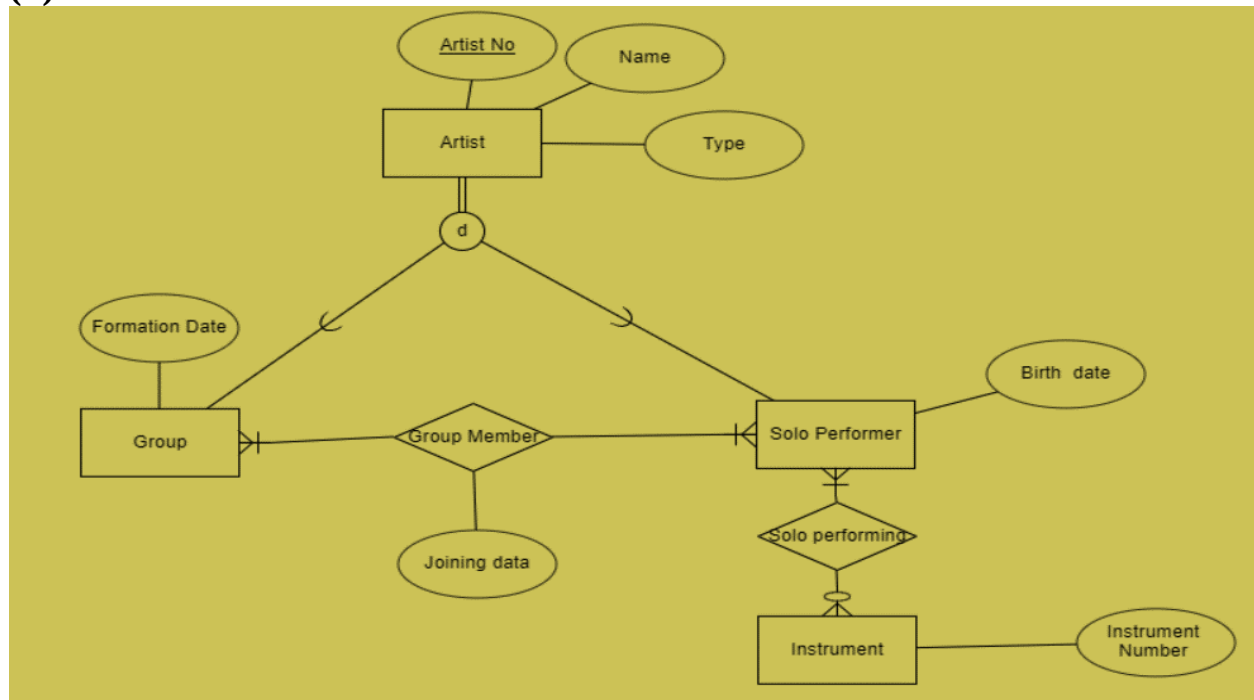
Reservation(Reservation-Number, Check-In, Check-out, Room-ID, Customer-ID)

Customer(Customer-ID, Customer-Name, City, State, Post-Code)



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(b)



Solution

Artist (Artist-No, Name, Type)

Group(Artist-No, Formation-Date)

Sole-Performer(Artist-No, Birth-date)

Instrument(Instrument-Number, Name)

Sole-Performing(Artist-No, Instrument-No)